

Question ID: c74df824

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Ratios, rates, proportional relationships, and units	Hard

Question

An area of **46.00** square nautical miles is equivalent to k square kilometers. To the nearest tenth, what is the value of k ?
(**1 nautical mile = 1.852 kilometers**)

Correct Answer: 157.8, 789/5

Rationale

The correct answer is **157.8**. It's given that an area of **46.00** square nautical miles is equivalent to k square kilometers. Since **1** nautical mile is equal to **1.852** kilometers, it follows that an area of **1** square nautical mile is equivalent to $(1.852)^2$, or **3.429904**, square kilometers. Multiplying **46.00** by **3.429904** yields **157.775584** square kilometers. Rounding this value to the nearest tenth yields **157.8**. Therefore, the value of k is **157.8**.